Sri Chaitanya School
Techno Curriculum

The right mentor for IIT (JEE), Medical, Olympiad & all other Competitive exams

STATEWIDE GRAND TEST - I

Max.Marks : 720 M

Class : M_2 MPL **NEET MODEL** **Time: 3 Hrs**

Subjects : Physics, Chemistry, Botany & Zoology. **Date : 23.04.2025**

SYLLABUS

Physics	:	Oscillations Mechanical properties of fluids : Upto Archemede's principale and its applications
Chemistry	:	Ionic equilibrium Thermodynamics
Botany	:	Biomolecule (Complete) Plant Growth and Development (Complete)
Zoology	:	Neural control and coordination (Complete) Chemical coordination and integration (Complete)

Instructions :

i) Each question consists of Four options with single correct answer.

ii) The test is of 3 hours duration and this question paper contains 180 questions.

iii) Each question carries ***4 marks****. For each correct response, the candidate will get 4 marks.*

For each incorrect response, one mark will be deducted *from the total scores. The maximum marks are 720.*

iv) Use Blue / Black Ball Point Pen only for writing particulars on OMR page / marking responses..

Name of the Student ..... Roll No .....

PHYSICS

1. The displacement of a simple harmonic oscillator is given by $y = 4\cos(2\pi t + \pi/4)m$. Then velocity of the oscillator at t = 2 seconds is

1) $4\sqrt{2}\pi ms^{-1}$ 2) $\frac{4\pi}{\sqrt{2}} ms^{-1}$ 3) $\frac{\sqrt{2}\pi}{4} ms^{-1}$ 4) $\frac{4\sqrt{2}}{\pi} ms^{-1}$

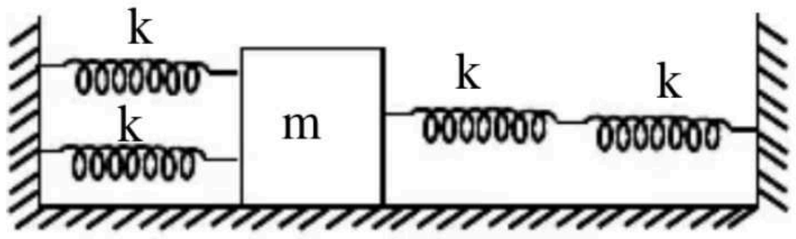
2. A flat horizontal board moves in SHM up and down. A small coin is placed on it. The maximum permissible amplitude of the board such that the coin does not separate from the board if its frequency of oscillation is f

1) $\frac{g}{4\pi^2 f^2}$ 2) $\frac{g}{2\pi^2 f^2}$ 3) $\frac{2g}{\pi^2 f^2}$ 4) $\frac{4g}{\pi^2 f^2}$

3. A particle of mass 0.1 kg is executing SHM of amplitude 0.1m. When the particle passes through the mean position, its kinetic energy is 8×10^{-3} J. Then the equation of its motion when the initial phase is 45° is

1) $0.2\sin(4t + \pi/4)$ 2) $0.1\sin(4t + \pi/4)$ 3) $0.1\sin(2t + \pi/4)$ 4) $0.1\sin(t + \pi/4)$

4. Four massless springs each of force contant K are attached to a mass m kept on a smooth plane as shwon. If the mass is displaced in the horizontal direction and released, it frequency of oscillation is



1) $\frac{1}{2\pi}\sqrt{\frac{K}{4m}}$ 2) $\frac{1}{2\pi}\sqrt{\frac{4K}{m}}$ 3) $\frac{1}{2\pi}\sqrt{\frac{2K}{5m}}$ 4) $\frac{1}{2\pi}\sqrt{\frac{5K}{2m}}$

5. Two simple pendulums of lengths 100cm and 1600cm are in phase at the mean position at a certain instant of time. If T is the time period of first pendulum, then the minimum time after which they will be in phase again is

1) $\frac{3T}{4}$ 2) $\frac{4T}{3}$ 3) 3T 4) 4T

6. A pendulum clock of length l is moving up with deceleration g/3. Then the number of oscillations made by that pendulum in 2 seconds is

1) $\frac{1}{\pi}\sqrt{\frac{2g}{3l}}$ 2) $\frac{1}{2\pi}\sqrt{\frac{2g}{3l}}$ 3) $\frac{1}{\pi}\sqrt{\frac{3g}{2l}}$ 4) $\frac{1}{2\pi}\sqrt{\frac{3g}{2l}}$

7. If two simple harmonic motions of same amplitude and frequency with a phase difference $\pi/2$ are superimposed, the resulting motion is

1) Linear 2) Circular 3) Simple harmonic 4) Rolling

8. The seconds pendulum is taken from earth to moon, to keep the time period constant

1) Length of the seconds pendulum should be reduced
2) The length of the seconds pendulum should be raised
3) The amplitude should increase
4) The amplitude should decrease

9. When the length of a simple pendulum is increased by 36cm, its time period of oscillation is found to increase by 25%. The initial length of the simple pendulum is (Acceleration due to gravity = 9.8 ms^{-2})
1) 36 cm 2) 25 cm 3) 64 cm 4) 46 cm

10. The potential energy of a particle of mass 5mg executing simple harmonic motion along x-axis is given by $V(x) = (1.23) + (4.84)x^2$, where V(x) is in joules and x is in metres. The frequency of oscillation of the particle is Hz.
1) 77 2) 7 3) 22 4) 2.2

11. The time period of a simple pendulum of length 'L' is T_1. Time period of a uniform rod of same length 'L' suspended from one end and oscillationg in a vertical plane is T_2. Amplitude of oscillation is small in both cases. Then, $\frac{T_1}{T_2}$ is
1) $\sqrt{\frac{4}{3}}$ 2) 1 3) $\sqrt{\frac{3}{2}}$ 4) $\sqrt{\frac{2}{3}}$

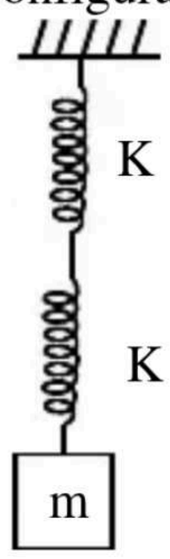
12. An object undergoing simple harmonic motion takes 0.5s to travel from one point of zero velocity to the next such point. The angular frequency of the motion is
1) $\pi\ rad\ s^{-1}$ 2) $2\pi\ rad\ s^{-1}$ 3) $3\pi\ rad\ s^{-1}$ 4) $\frac{\pi}{2} rad\ s^{-1}$

13. A body starting at t = 0 from origin and oscillates simple harmonically with a period of 4s. After what time will its kinetic energy be 75% of its total energy
1) 1/2s 2) 1/3s 3) 1/4s 4) 1s

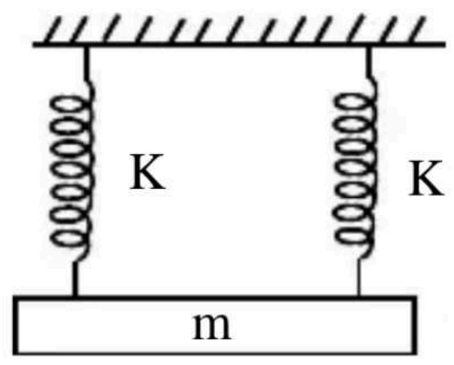
14. The displacement of a particle in simple harmonic motion (SHM) is given by $y = \sqrt{3}\pi \sin\left(\frac{100}{\pi}t + \frac{\pi}{4}\right)$. What will be the displacement of the particle from the mean position when its kinetic energy is eight times that of its potential energy ?
1) $\sqrt{\frac{\pi}{3}}$ 2) $\sqrt{\frac{2\pi}{3}}$ 3) $\sqrt{\pi}$ 4) $\sqrt{3\pi}$

15. A mass 'M' is suspended from a spring of negligible mass. The spring is pulled down a little and then released so that the mass executes simple harmonic oscillations of a time period T. If the mass attached is increased by 'm', the time period was observed to become 9T/4. Find the ratio m : M.
1) 9 : 16 2) 65 : 16 3) 16 : 9 4) 16 : 65

16. Two identical springs are connected to mass m as shown (K = spring constatn). If the period of the configuration in (i) is 2s. the period of the configuration (ii) is :
1) $\sqrt{2}$ s
2) 1 s
3) $\frac{1}{\sqrt{2}}$ s
4) $2\sqrt{2}$ s



(i)



(ii)

17. The average energy in one period of a simple harmonic motion is

1) $\frac{1}{2}m\omega^2a^2$ 2) $m\omega^2a^2$ 3) $\frac{3}{2}m\omega^2a^2$ 4) $2m^2\omega^2a^2$

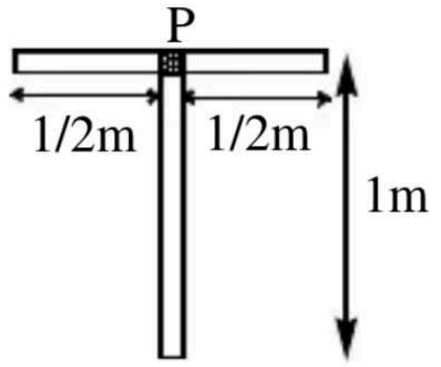
18. A physical pendulum consists of two thin rods of equal mass, each of length 1 meter. The rods are joined together at point P as shown in the figure below. The angular frequency of the pendulum about pont P (which is at the center of horizontal rod) is (Assume g = 10 m/s^2)

1) $2\sqrt{3}\,rad/s$

2) $3\sqrt{2}\,rad/s$

3) $4\,rad/s$

4) $3\sqrt{5}\,rad/s$



19. A particle is exhibiting simple harmonic motion has its displacement (x) and velocity (v) related as $4v^2 = 25 - x^2$. The time period of SHM is

1) π sec 2) 2π sec 3) 3π sec 4) 4π sec

20. For a particle executing SHM, determine the ratio of average acceleraton of the particle between extreme position and equilibrium position w.r.t the maximum acceleration.

1) $\frac{4}{\pi}$ 2) $\frac{2}{\pi}$ 3) $\frac{1}{\pi}$ 4) $\frac{1}{2\pi}$

21. A spring is stretched 0.40 m when a mass of 0.6 kg is suspended from it. The period of oscilations of the spring loaded by 225 gm and put to oscillations close to _____ ($g = 10\ ms^{-2}$)

1) 11 s 2) 48.6s 3) 0.82 s 4) 4.86 s

22. A simple pendulum is vibratring in an evacuated chamber. It will oscillate with

1) Constant amplitude
2) Decreasing amplitude
3) Increasing amplitude
4) Amplitude, the variation of which cannot be predicted

23. Ratio of maximum acceleration to the maximum velocity of a simple harmonic oscillator is

1) ω 2) $\frac{\omega}{2}$ 3) ω^2 4) 2ω

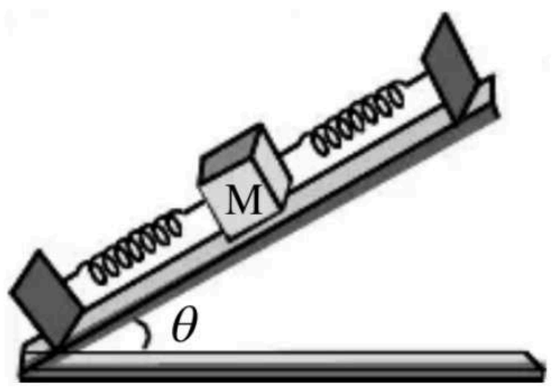
24. A simple pendulum, suspended from the celling of a lift, has a period of oscillatoin T when the lift is at rest. If the lift starts moving upwards with an acceleration a = 3g, then the new period will be

1) 2T 2) 4T 3) $\frac{T}{3}$ 4) $\frac{T}{2}$

25. On a smooth inclined plane, a mass M is attached between two massless springs of force constant k each, as shown in the figure. The other ends of the springs are fixed to firm supports. The period of oscillation of the mass M is

1) $2\pi\left(\frac{M}{2k}\right)^{1/2}$ 2) $2\pi\left(\frac{2M}{k}\right)^{1/2}$

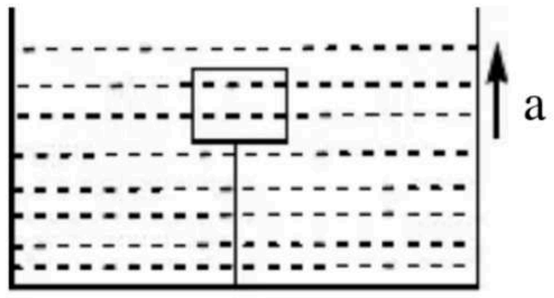
3) $2\pi\left(\frac{Mg\sin\theta}{2k}\right)$ 4) $2\pi\left(\frac{2Mg}{k}\right)^{1/2}$



26. A particle executes simple harmonic motion between $x=-A$ and $x=+A$. If it takes a time T_1 to go from $x=0$ to $x=\frac{A}{2}$ and T_2 to go from $x=\frac{A}{2}$ to $x=A$. Then

1) $T_1 < T_2$ 2) $T_1 > T_2$ 3) $T_1 = T_2$ 4) $T_1 = 2T_2$

27. The tension in a string holding a solid block below the surface of a liquid (density greater than that of solid) is T_0 when the system is at rest. If the system has upward acceleration 'a', tension in the string would be



1) T_0 2) $T_0(a-g)$ 3) $T_0\left(1+\frac{a}{g}\right)$ 4) $T_0\left(1-\frac{a}{g}\right)$

28. A cube of wood supporting a 200gm mass just floats in the water. When the mass is removed, the cube rises by 2cm. Then length of side of the cube is

1) 8 cm 2) 12 cm 3) 6 cm 4) 10 cm

29. When a body is weighed in a liquid the apparent loss in its weight is equal to

a) The up thrust of liquid on the body
b) Weight of the liquid displaced by the body
c) The difference in weights of the body in air and liquid
d) Weight of water displaced by the body

1) a, b, c are correct 2) a, b are correct 3) a, b, d are correct 4) b, d are correct

30. A block of wood floats in a liquid with four-fifths of its volume submerged. If the relative density of wood is 0.8, what is the density of the liquid in units of kg m^{-3} ?

1) 750 2) 1000 3) 1250 4) 1500

31. A cubical block is floating in a liquid with half of its volume immersed in the liquid. When the whole system accelerates upwards with acceleration of $\frac{g}{3}$, the fraction of volume immersed in the liquid will

1) $\frac{1}{2}$ 2) $\frac{3}{8}$ 3) $\frac{2}{3}$ 4) $\frac{3}{4}$

32. A vessel contains oil of density 0.8 gm cm^{-3} floating over mercury of density 13.6 gm cm^{-3}. A homogeneous sphere floats with half its volume immersed in mercury and the other half in oil. The density of the sphere in gm cm^{-3} is

1) 3.3 2) 6.4 3) 7.2 4) 12.8

33. An ice bag is floating in sea water. The density of ice is 0.92 gm cm^{-3} and that of sea water is 1.03 gm cm^{-3}. The percentage of ice bag below surface of water is

1) 3% 2) 11% 3) 89% 4) 92%

34. A metal sphere of volume V, having a cavity inside it, floats on water completely submerged. If the relative density of the metal is 1.5, the volume of the cavity is

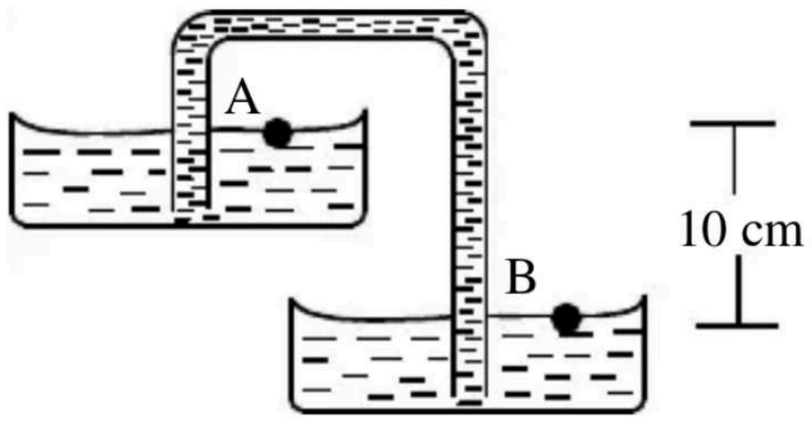
1) $\frac{V}{4}$ 2) $\frac{V}{3}$ 3) $\frac{V}{2}$ 4) $\frac{2V}{3}$

35. The three vessels shown in figure have same base area. Equal volumes of a liquid are poured in the three vessels. The force on the base will be

1) Maximum in vessels A
2) Maximum in vessels B
3) Maximum in vessels C
4) Equal in all the vessesls



36. Figure shows a siphon. The liquid shown is water. The pressure difference $P_B - P_A$ between the points A and B is



1) 400 N/m^2 2) 3000 N/m^2 3) 1000 N/m^2 4) Zero

37. **Assertion (A) :** A solid sphere and a hollow sphere of same material are floating in a liquid. Radius of both the spheres are same. Percentage of volume immersed of both the spheres will not be same.

Reason (R) : Upthrust depends on volume of liquid displaced.

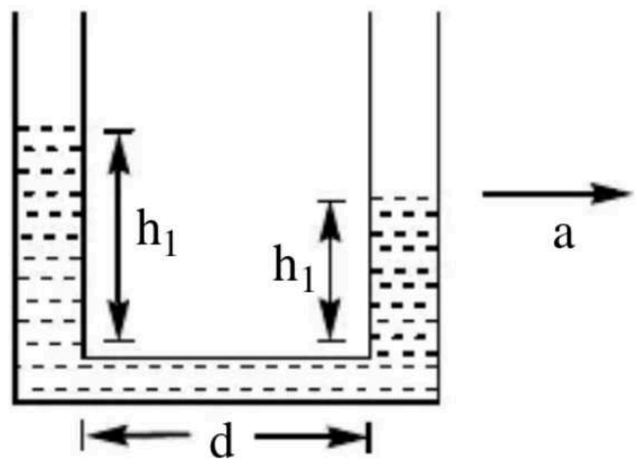
1) Both A and R are true and R is the correct explanation of A
2) Both A and R are true and R is not the correct explanation of A
3) A is true but R is false
4) A is false but R is true

38. The density of the atmosphere at sea level is 1.29 kg/m^3. Assume that it does not change with altitude. Then how high would the atmosphere extend ?

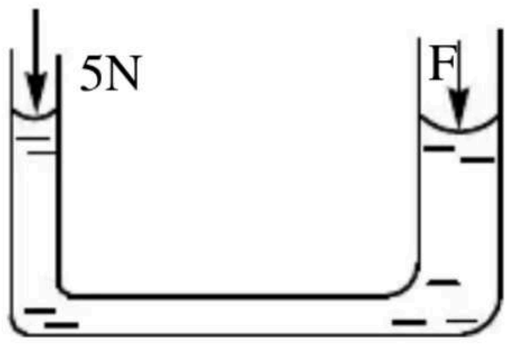
1) 3 km 2) 7 km 3) 8 km 4) 9 km

39. Figure shows a U-tube of uniform cross sectional area A accelerated with acceleration a as sown. If d is the separation between the limbs, then the difference in the levels of the liquid in the U-tube is

1) $\frac{ad}{g}$
2) $\frac{g}{ad}$
3) adg
4) ad + g



40. The area of cross section of the two arms of a hydraulic press are 1 cm^2 and 10 cm^2 respectively (figure). A force of 5 N is applied on the water in the thinner arm. What force should be applied on the water in the thicker arm so that the water may remain in equilibrium ?

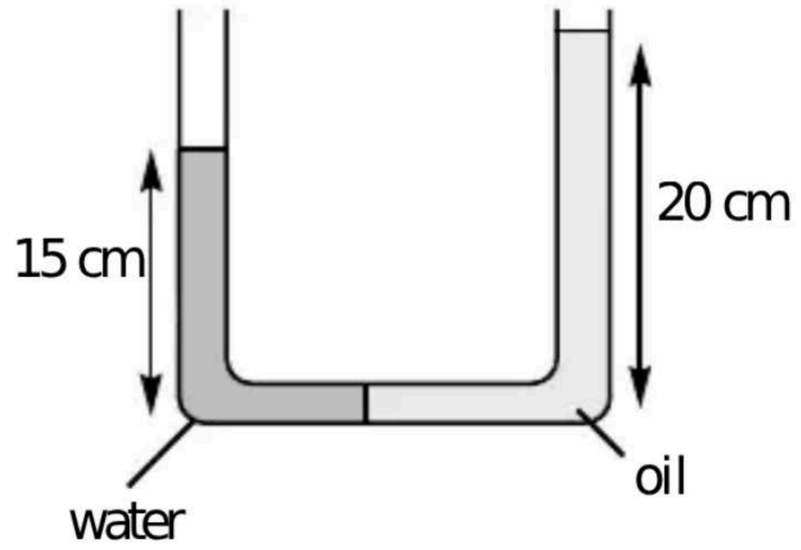


1) 50 N 2) 60 N 3) 70 N 4) 80 N

41. Two solids A and B float in water. It is observed that A floats with half of its volume immersed and B floats with 2/3 of its volume immersed. The ratio of their densities is
1) 4 : 3 2) 3 : 4 3) 2 : 3 4) 3 : 2

42. A block of solid insoluble in water wieghs 24 gm in air and 21 gm when completely immersed in water. Its weight when completely immersed in liquid of specific gravity 1.1 is
1) 20.7 gm 2) 27.3 gm 3) 24 gm 4) 3.3 gm

43. In a u-tube as shown in figure, water and oil are in the left side and right side of the tube respectively. The heights form the bottom for water and oil columns are 15 cm and 20 cm respectively. The density of the oil is : [take $\rho_{water} = 1000 kg/m^3$]



1) 1200 kg/m^3 2) 750 kg/m^3 3) 1000 kg/m^3 4) 1333 kg/m^3

44. A metal sphere weighting 60 gm in air is suspended by a thread in an oil of specific gravity 0.6. If the tension in the thread is 55gmwt , the specific gravity of the metal sphere is
1) 9.4 2) 4.5 3) 6.3 4) 7.2

45. Two spheres of volume 250 cc each but of relative densities 0.8 and 1.2 are connected by a string and the combination is immersed in a liquid find tension in the string (g = 10 m/s^2)
1) 0.5 N 2) 1 N 3) 3 N 4) 2.5 N

CHEMISTRY

46. The number of H_3O^+ ions present in 10 ml of water at 25°C is
1) 6.023×10^{-14} 2) 6.023×10^{14} 3) 6.023×10^{-19} 4) 6.023×10^{19}

47. The $[OH^-]$ of 0.005 M H_2SO_4 is
1) 2×10^{-12} M 2) 5×10^{-3} M 3) 10^{-2} M 4) 10^{-12} M

48. The enthalpy change of a reaction does not depend on
1) The state of reactants and products 2) Nature of reactants and products
3) Different intermediate reaction 4) Initial and final enthalpy change of a reaction

49. 1g sample of NH_4NO_3 is decomposed in a bomb calorimeter .The temperature of the calorimeter increases by 6.12K.The heat capacity of the system is 12.3J/g / deg. What is the molar heat of decomposition for NH_4NO_3?
1) – 7.53kJ/ mol 2) – 398.1kJ / mol 3) – 16.1kJ /mol 4) – 6.2kJ/ mol

50. Which of the following pair of solutions does not form a buffer solution ?
1) NaH_2PO_4 and Na_2HPO_4 2) H_2CO_3 and $NaHCO_3$
3) NH_4OH and NH_4Cl 4) KOH and K_2SO_4

51. The solubility of calcium phosphate in water is x mol L^{-1} at 25°C. Its solubility product is equal to
1) 108 x^2 2) 36x^3 3) 36 x^5 4) 108 x^5

52. The enthalpy of vaporization of liquid water using the data
$H_2(g) + 1/2 0_2(g) \rightarrow H_2O(l);\ \Delta H = -285.77 kJ/mole$
$H_2(g) + 1/2 0_2(g) \rightarrow H_2O(g);\ \Delta H = -241.84 kJ/mole$ is
1) + 43.93 kJ/ mol 2) – 43.93kJ/ mole 3) + 527.61 kJ/mol 4) – 527.61kJ/mol

53. If order to decompose 9g water , 142.5kJ heat is required .Hence the enthalpy of formation of water is
1) + 285 kJ 2) – 285 kJ 3) + 142.6kJ 4) – 142.5kJ

54. For a weak acid, the concentration of H^+ ions is given by
1) $\sqrt{K_a.C}$ 2) K_a / C 3) $\sqrt{K_a / C}$ 4) $\sqrt{C / K_a}$

55. For $H_3PO_4, H_3PO_4 \rightleftharpoons H_2PO_4^- + H^+(K_1);\ H_2PO_4^- \rightleftharpoons HPO_4^{2-} + H^+(K_2);\ HPO_4^{2-} \rightleftharpoons PO_4^{3-} + H^+(K_3)$, then
1) $K_1 > K_2 > K_3$ 2) $K_1 < K_2 < K_3$ 3) $K_1 < K_2 > K_3$ 4) $K_1.K_2.K_3 = K_W$

56. **Assertion (A) :** Aqueous solution of Na_2CO_3 shows pH > 7.
Reason (R) : Salt of strong base and weak acid undergoes anionic hydrolysis.
1) Both A and R are true and R is the correct explanation of A
2) Both A and R are true and R is not the correct explanation of A
3) A is true but R is false 4) A is false but R is true

57. Ionic product of water depends on
1) Volume of the water 2) Amount of salt in water
3) Temperature 4) All the above

58. Which of the following is relatively stronger acid ?
1) H_2S 2) HCl 3) HF 4) CH_3COOH

59. Which of the following species acts as Bronsted base but not as acid ?
1) CH_3COOH 2) HCO_3^- 3) $H_2PO_2^-$ 4) $H_2PO_3^-$

60. Which of the following is a buffer solution ?
1) 500 mL of 0.1 N CH_3COOH + 500 mL of 0.1 N NaOH
2) 500 mL of 0.1 N CH_3COOH + 500 mL of 0.1 N HCl
3) 500 mL of 0.1 N CH_3COOH + 500 mL of 0.2 N NaOH
4) 500 mL of 0.2 N CH_3COOH + 500 mL of 0.1 N NaOH

61. Hess's law states that
1) The standard enthalpy of an overall reaction is the sum of the enthalpy changes in individual reactions
2) Enthalpy of formation of a compound is same as the enthalpy of decomposition of the compound into constituent elements, but with opposite sign
3) At constant temperature, the pressure of a gas is inversely proportional to its volume
4) The mass of a gas dissolved per litre of a solvent is proportional to the pressure of the gas in equilibrium with the solution

62. Under the same conditions, how many mL of 1M KOH and 0.5M H_2SO_4 solutions, respectively, when mixed to form a total volume of 100 mL produces the highest rise in temperature?
1) 67, 33 2) 33, 67 3) 40, 60 4) 50,50

63. If at 298K, the bond energies C – H, C – C, C = C and H – H bonds are respectively 414,347,615 and 435 kJ mol^{-1}, the value of enthalpy change for the reaction
$H_2C = CH_2(g) + H_2(g) \rightarrow H_3C - CH_3(g)$ at 298 K will be
1) –125 kJ 2) +250 kJ 3) –250 kJ 4) + 125 kJS

64. Which one of the following has ΔS^0 positive

1) $CaO(s)+CO_2(g) \rightleftharpoons CaCO_3(s)$ 2) $NaCl(aq) \rightleftharpoons NaCl(s)$

3) $NaNO_3(s) + H_2O(1) \rightleftharpoons Na^+(aq) + NO_3^-(aq)$ 4) $N_2(g) + 3H_2(g) \rightleftharpoons 2NH_3(g)$

65. For a reaction at 25^0C, enthalpy change (ΔH) and entropy change (ΔS) are – 11.7 kJ mol^{-1} and –105J mol^{-1} K^{-1} respectively .The reaction is

1) Spontaneous at this temperature 2) non - spontaneous at this temperature
3) At 100^0C 4) None of these

66. Which has the highest entropy per mol of the substance?

1) H_2 at 25^0C at 1 atm 2) H_2 at STP
3) H_2 at 100K at 1 atm 4) H_2 at 0 K at 1 atm

67. Consider following reactions

I : $N_2 + O_2 \rightarrow 2NO \quad \Delta H = x_1$ II : $2CO + O_2 \rightarrow 2CO_2 \quad \Delta H = x_2$

III : $2H_2O + O_2 \rightarrow 2H_2O_2 \quad \Delta H = x_3$ IV : $PCl_3 + Cl_2 \rightarrow PCl_5 \quad \Delta H = x_4$

In which case (s) , $\Delta H_f = \frac{\Delta H}{2}$?

1) I,II,III 2) IV 3) II,III 4) I

68. At 25°C, for an acid

1) $[H^+] > 10^{-7}$ M 2) $[OH^-] < 10^{-7}$M 3) $[H^+] = 10^{-7}$ 4) 1 and 2

69. What is the pH of a solution obtained by dissolving 0.0005 mole of the strong electrolyte, calcium hydroxide, $Ca(OH)_2$ to form 100 ml of a saturated solution (aqueous)?
($K_w = 1.0 \times 10^{-14}$ $mole^2$ $litre^{-2}$)

1) 9.8 2) 11.7 3) 12.0 4) 3.0

70. K_{sp} of AgCl is 1×10^{-10}. Its solubility in 0.1 M KNO_3 will be :

1) 10^{-5} moles/litre 2) $> 10^{-5}$ moles/ litre 3) $< 10^{-5}$ moles/ litre 4) None of these

71. A system undergoes a process in which $\Delta U = +300J$ while absorbing 400 J of heat energy and undergoing an expansion against 0.5 bar. What is the change in the volume (in L) ?

1) 4 2) 5 3) 2 4) 3

72. When 0.1 mole of an acid is added to 2 lit of a buffer solution, the pH of the buffer decreases by 0.5. The buffer capacity of the solution is

1) 0.6 2) 0.4 3) 0.2 4) 0.1

73. Bond - energies of N ≡ N bond , H– H bond and N – H bond are respectively x_1,x_2 and x_3 kJ mol^{-1}. Hence, ΔH_f^0 (NH_3) is

1) $x_1 + 3x_2 - x_3$ 2) $x_1 + x_2 - x_3$ 3) $\frac{x_1}{2} + \frac{3}{2}x_2 - 3x_3$ 4) $x_1 + 3x_2 - 6x_3$

74. Following reaction occurs at 25^0C : $2NO(g, 1 \times 10^{-5}$ atm) +

Cl_2 (g , 1×10^{-2} atm) $\rightleftharpoons 2NOCl$ (g, 1×10^{-2} atm) $\Delta G^0 is$

1) – 45.65 kJ 2) – 28.53kJ 3) – 22.83kJ 4) – 57. 06 kJ

75. An endothermic reaction is spontaneous if :

1) $\Delta H > T\Delta S$ 2) $\Delta H < T\Delta S$ 3) $\Delta H = T\Delta S$ 4) $T\Delta S = 0$

76. One lit. of aqueous solution of H_2SO_4 contains 4.9 g of the acid. 100 ml of this solution is taken in a 1 lit. Flask and diluted with water upto the mark. The pH of the dilute solution is
1) 2.0 2) 2.301 3) 1.699 4) 3.699

77. Let the solubilities of AgC*l* in H_2O, 0.01M $CaCl_2$, 0.01M NaC*l* and 0.05M $AgNO_3$ be S_1, S_2, S_3 and S_4 respectively. What is the correct relationship between these quantities ?
1) $S_1 > S_2 > S_3 > S_4$ 2) $S_1 > S_2 = S_3 > S_4$ 3) $S_1 > S_3 > S_2 > S_4$ 4) $S_4 > S_2 > S_3 > S_1$

78. The pK_a of weak acid HA is 4.80 and the pK_b of a weak base BOH is 4.78. The pH of an aqueous solution of corresponding salt BA will be :
1) 9.22 2) 9.58 3) 4.79 4) 7.01

79. Hydrolysis constants of two salts KA and KB of weak acids HA and HB are 10^{-8} and 10^{-6}. If the dissociation constant of third acid HC is 10^{-2}. The order of acidic strengths of three acids will be
1) HA > HB > HC 2) HB > HA > HC 3) HC > HA > HB 4) HA = HB = HC

80. $\Delta H_f^{\odot}$ 298 K of methanol is given by the chemical equation

1) $CH_4(g)+\frac{1}{2}O_2(g)\rightarrow CH_3OH(g)$

2) $C(graphite)+\frac{1}{2}O_2(g)+2H_2(g)\rightarrow CH_3OH(l)$

3) $C(diamond)+\frac{1}{2}O_2(g)+2H_2(g)\rightarrow CH_3OH(l)$

4) $CO(g)+2H_2(g)\rightarrow CH_3OH(l)$

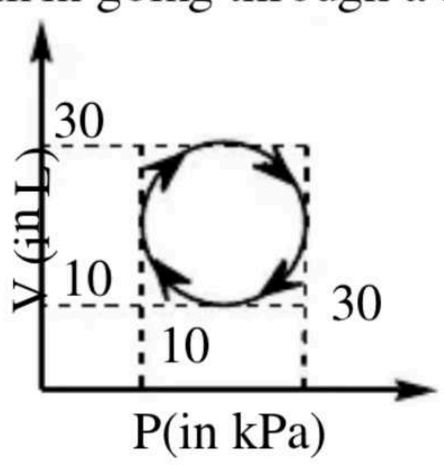
81. Heat absorbed by a system in going through a cyclic process shown in figure is :

1) $10^7\pi J$

2) $10^6\pi J$

3) $10^2\pi J$

4) $10^4\pi J$



82. Enthalpy of neutralisation of the reaction between CH_3COOH (aq) and NaOH(aq) is – 13.2 kcal Eq^{-1} and that of the reaction between H_2SO_4 (aq) and (KOH) (aq) is – 13.7 kcal Eq^{-1}. The enthalpy of dissociation of CH_3COOH(aq) is
1) – 0.5 kcal Eq^{-1} 2) + 0.5kcal Eq^{-1} 3) – 26.9 kcal Eq^{-1} 4) + 13.45kcal Eq^{-1}

83. Given $\Delta_i H^{\odot}$ (HCN) = 45.2kJ $mol^{-1}$and $\Delta_i H^{\odot}$ (CH_3 COOH) = 2.1kJ mol^{-1}. Which one of the following facts is true ?

1) $pK_a(HCN) = pK_a(CH_3COOH)$

2) $pK_a(HCN) > pK_a(CH_3COOH)$

3) $pK_a(HCN) < pK_a(CH_3COOH)$

4) $pK_a(HCN) = (45.17/2.07)pK_a(CH_3COOH)$

84. Adiabatic reversible expansion of a gas is represented by :
1) $\left(\frac{T_1}{T_2}\right)^{\gamma}=\left(\frac{P_2}{P_1}\right)^{1-\gamma}$ 2) $\left(\frac{T_1}{T_2}\right)^{\gamma}=\left(\frac{P_1}{P_2}\right)^{1-\gamma}$ 3) $\left(\frac{T_1}{T_2}\right)^{\gamma}=\left(\frac{P_2}{P_1}\right)^{\gamma-1}$ 4) all are correct

85. Melting point of a solid is x K and its latent heat of fusion is 600 cal mol^{-1} .The entropy change for fusion of 1 mol solid is 2 cal mol^{-1} K^{-1}.The value of x will be
1) 100K 2) 200 K 3) 300 K 4) 400 K

86. In an equilibrium reaction, for which $\Delta G^\circ = 0$, the equilibrium constant K should be equal to:
1) zero 2) 10 3) 1 4) 2

87. For a spontaneous reaction ΔG, equilibrium K and E°_{cell} will be , respectively
1) $-ve, > 1, +ve$ 2) $-ve, < 1, -ve$ 3) $+ve, > 1, -ve$ 4) $-ve, > 1, -ve$

88. For the reaction at 298 K, 2A + B $\rightarrow$C $\Delta H = 100 kcal$ and $\Delta S = 0.020 kcal K^{-1}$. If ΔH and ΔS are assumed to be constant over the temperature range, at what temperature will the reaction become spontaneous ?
1) 1000K 2) 3500 K 3) 6000 K 4) 1500 K

89. Which among the following statements is/ are incorrect ?
1) $pH = -\log_{10}(H_3O^+)$
2) pH decreases with increase of temperature
3) pH cannot be zero, negative or more than 14
4) If a solution is diluted ten times, its pH increased by 1

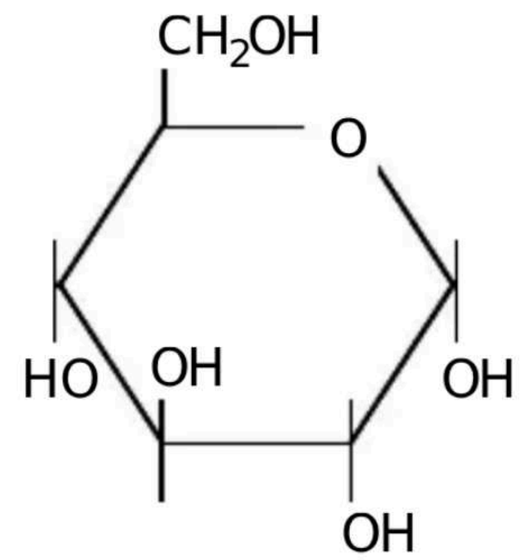
90. The degree of hydrolysis of which of the following salt is independent of the concentration of salt solution ?
1) CH_3COONa 2) NH_4Cl 3) CH_3COONH_4 4) NaCl

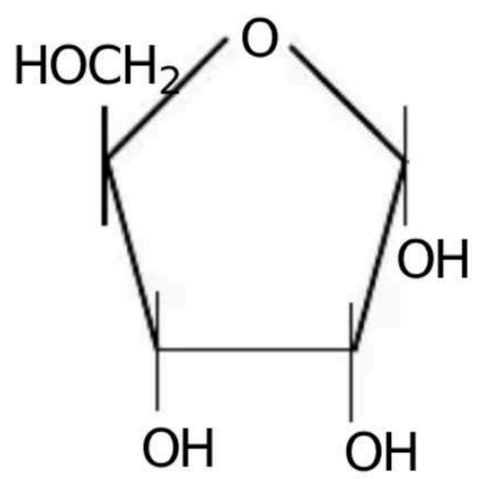
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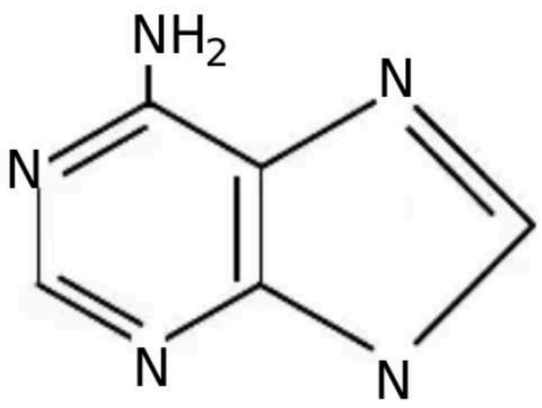
91. Which of the following is not a macromolecule ?
1) Nucleic acid 2) Polysaccharides 3) Lipids 4) Proteins

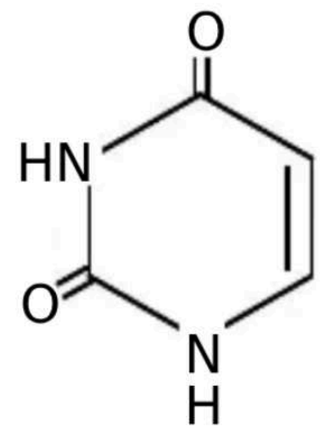
92. _____ of the total cell mass is formed by ions.
1) 1% 2) 2% 3) 3% 4) 4%

93. Which is the correct representation of ribose ?

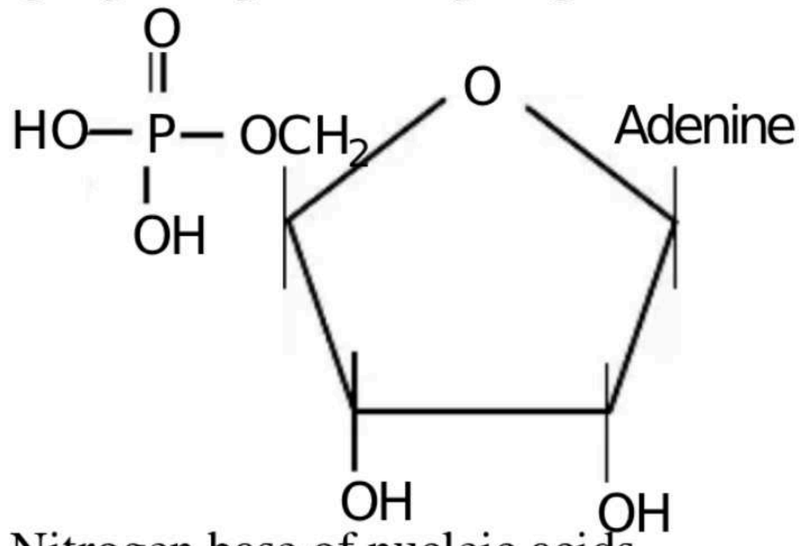
1) 

2) 

3) 

4) 

94. Select the correct matching regarding following diagram :



1) Purine – Nitrogen base of nucleic acids
2) Uracil – Exclusively found in RNA only
3) Adenylic acid – Monomer of both RNA and DNA
4) Cholesterol – Forms part of cell membrane

95. Protein is
1) Homopolymer 2) Heteropolymer 3) Monomer 4) All

96. Match the column :

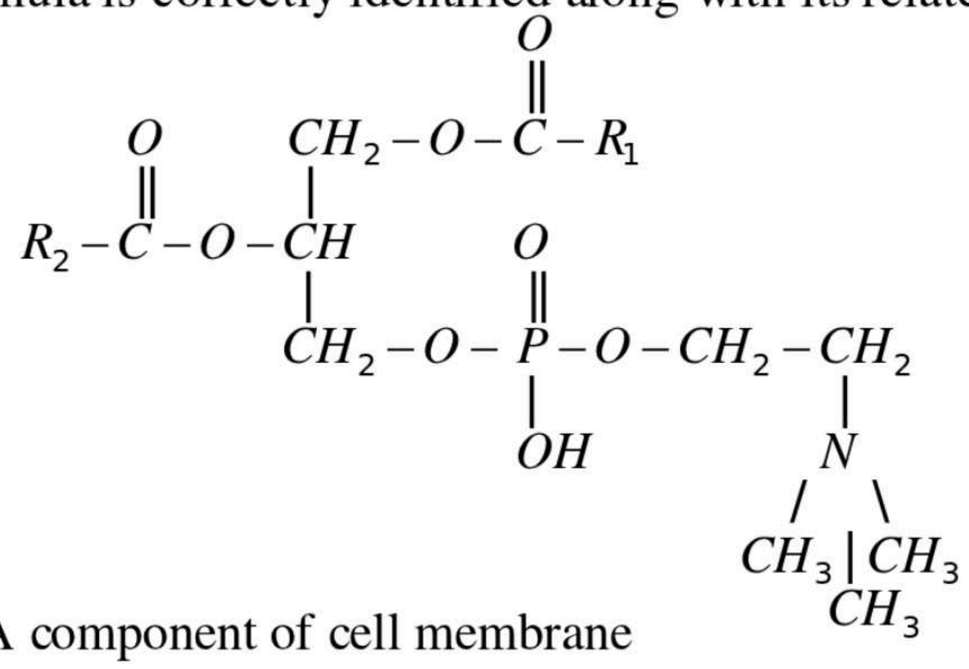
Column – I		**Column – II**
A) Collagen	–	1) Hormone
B) Trypsin	–	2) Enzyme
C) Insulin	–	3) Fight infectious agent
D) Receptro	–	4) Enable glucose transport into cells
E) GLUT – 4	–	5) Sensory reception (smell, taste hormone)
F) Antibody	–	6) Intercellular ground substance

1) A – 3, B – 2, C – 4, D – 1, E – 6, F – 5
2) A – 6, B – 2, C – 1, D – 5, E – 4, F – 3
3) A – 1, B – 4, C – 2, D – 5, E – 3, F – 6
4) A – 2, B – 4, C – 5, D – 3 , E – 1, F – 6

97. _____ of an enyme is a crevices or pocket into which substrate fit.
1) Inactive site 2) Active site 3) Allosteric site 4) Any of the above

98. The correct order of chemical composition of living tissues / cells in terms of % of the total of the total cellular mass is
1) Lipid > Ions > Carbohydrate > H_2O > Proteins > Nucleic acid
2) H_2O > Proteins > Carbohydrate > Nucleic acid > Lipid > Ions
3) H_2O > Proteins > Nucleic acid > Carbohydrate > Lipid > Ions
4) Nucleic acid > Proteins > H_2O > Carbohydrate > Ions > Lipid

99. Given structural formula is correctly identified along with its related funciton by which of the following options ?

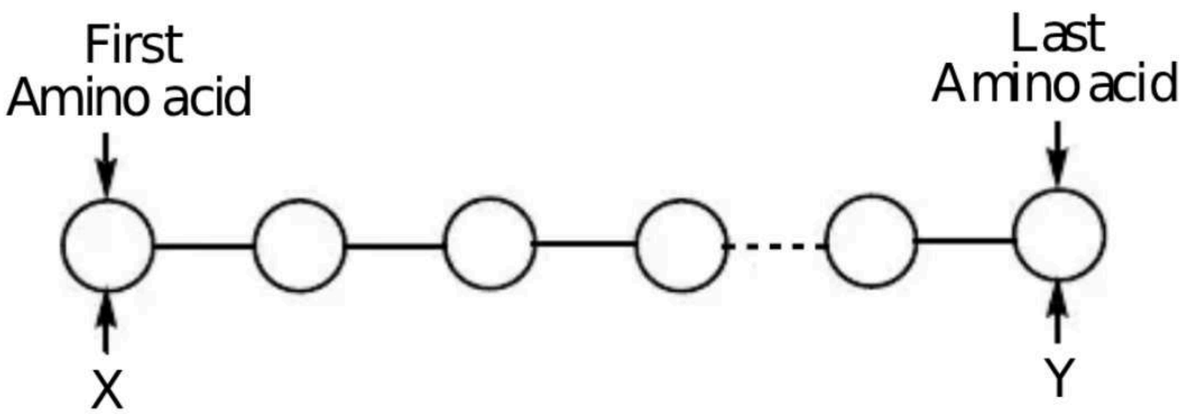


1) Lecithin – A component of cell membrane
2) Cholesterol – A component of animal cell membrane
3) Adenosine – A component of nucleic acids
4) Triglyceride – An energy source

100. The most abundant chemical in living organisms could be
1) Protein 2) Water 3) Sugar 4) Nucleic acid

101. Which is ture about the macromolecules or biomacromolecules ?
1) These are found in acid insoluble fraction or retentate
2) With the exception of lipids these are polymeric substance
3) They have molecular weight in the range of ten thoushands daltons and above (exception is lipid whose molecular weight do not exceed 800 Da)
4) All of the above

102. Study the given statements and select the correct answer.
i) Cellulose is a homopolymer of glucose
ii) Inulin is a homopolymer of fructose
iii) Starch gives blue black colour and glycogen gives red colour with iodine solution
iv) Cellulose gives no colour with iodine solution.
1) Statements i, ii and iii are correct 2) Statemetns i, ii and iv are correct
3) Statements ii and iii are correct 4) All statements are correct

103.


1) Both X and Y are N-terminal amino acid
2) X is N - terminal amino acid and Y is C - terminal amino acid
3) Both X and Y are C - terminal amono acid
4) X is C - terminal amino acid and Y is N - terminal acid

104. Rubber is
1) Primary metabolite 2) Secondary metabolite
3) Polymeric substance 4) 2 and 3

105. Match colums I with column II and select the correct option from the given codes.

Column – I (Category)	**Column – II (Secondary metabolites)**
A) Pigments	i) Concanavalin A
B) Terpenoides	ii) Monoterpenes, diterpenes
C) Alkaloids	iii) Morphine, codeine
D) Lectins	iv) Carotenoids, anthocyanins

1) A – iv, B – ii, C – iii, D – i 2) A – iv, B – iii, C – ii, D – i
3) A – i, B – iv, C – iii, D – ii 4) A – i, B – iii, C – ii, D – iv

106. Which is wrongly matched ?
1) ATP – Nucleotide derivative 2) Fungi – Chitin
3) Phospholipid – Plasma membrane 4) Enzyme – Lipopolysaccharide

107. Glycogen is a homopolymer made of
1) Glucose units 2) Galactose units 3) Ribose units 4) Amino acids

108. Adult human haemoglobin consists of
1) 2 subunits (α,α) 2) 2 subunits (β,β) 3) 4 subunits ($2\alpha,2\beta$) 4) 3 subunits ($2\beta,1\beta$)

109. Read the given statements and select the correct option.
Statement I : Low temperature destroys enzymes by causing their denaturation
Statement II : High temperature preserves the enzymes in their inactive stage.
1) Both statements I and II are correct
2) Statement I is correct but statement II is incorrect
3) Statement I is incorrect but statement II is correct
4) Both statements I and II are incorrect

110. Lysine is _____ amino acid
1) Acidic 2) Basic 3) Neutral 4) Aromatic

111. Inhibition of succinate dehydrogenase by malonate is an example of
1) Non – competitive inhibition 2) Competitive inhibition
3) Allosteric inhibition 4) Negative feedback

112. Enzymes that catalyse removal of groups from substrates by mechanisms other than hydrolysis, and addition of groups to double bonds, are called
1) Ligases 2) Lyases 3) Hydrolases 4) Dehydrogenases

113. Dihydroxyacetone-3-phosphate and glyceraldehyde-3-phosphate are interconvertible. The enzyme responsible for this interconversion belongs to the category of
1) Isomerases 2) Ligases 3) Lyases 4) Hydrolases

114. Which of the following is an example of plasticity ?
1) Heterophylly in cotton, coriander and larkspur
2) Heterophylly in buttercup
3) Bolting in beet and cabbage
4) Both 1 and 2

115. ABA acts as an antagonist to
1) IBA 2) NAA 3) IAA 4) GA

116. Gibberellin was first of all isolated from
1) Fungus 2) Lichen 3) Virus 4) Bacteria

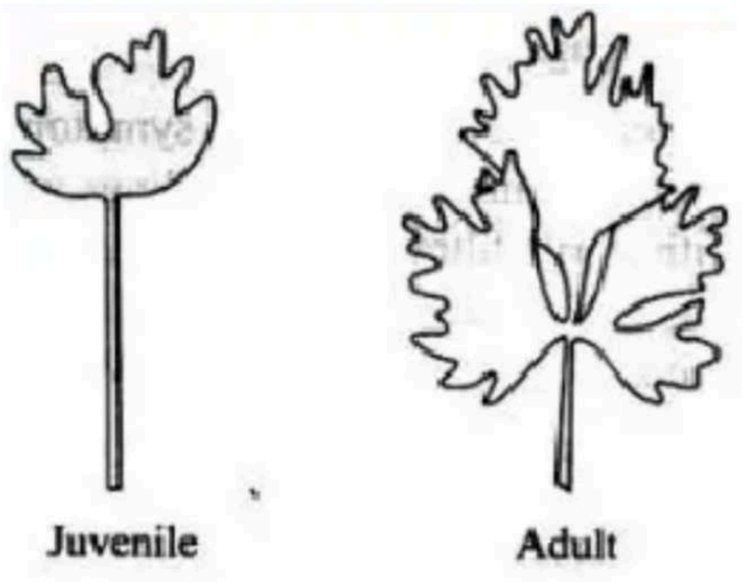
117. The development of an embryo displays
1) Geometric growth only
2) Arithmetic growth only
3) Arithmetic growth followed by geometric growth
4) Geometric growth followed by arithmetic growth

118. Re-differentiation results in fomration of
1) Secondary vascular tissue 2) Apical meristems
3) Intercalary meristems 4) All the above

119. During the development process in a plant cell, a mature cell is derived form a meristematic cell through
1) Differentation 2) Plasticity 3) Sensecence 4) Redifferentiation

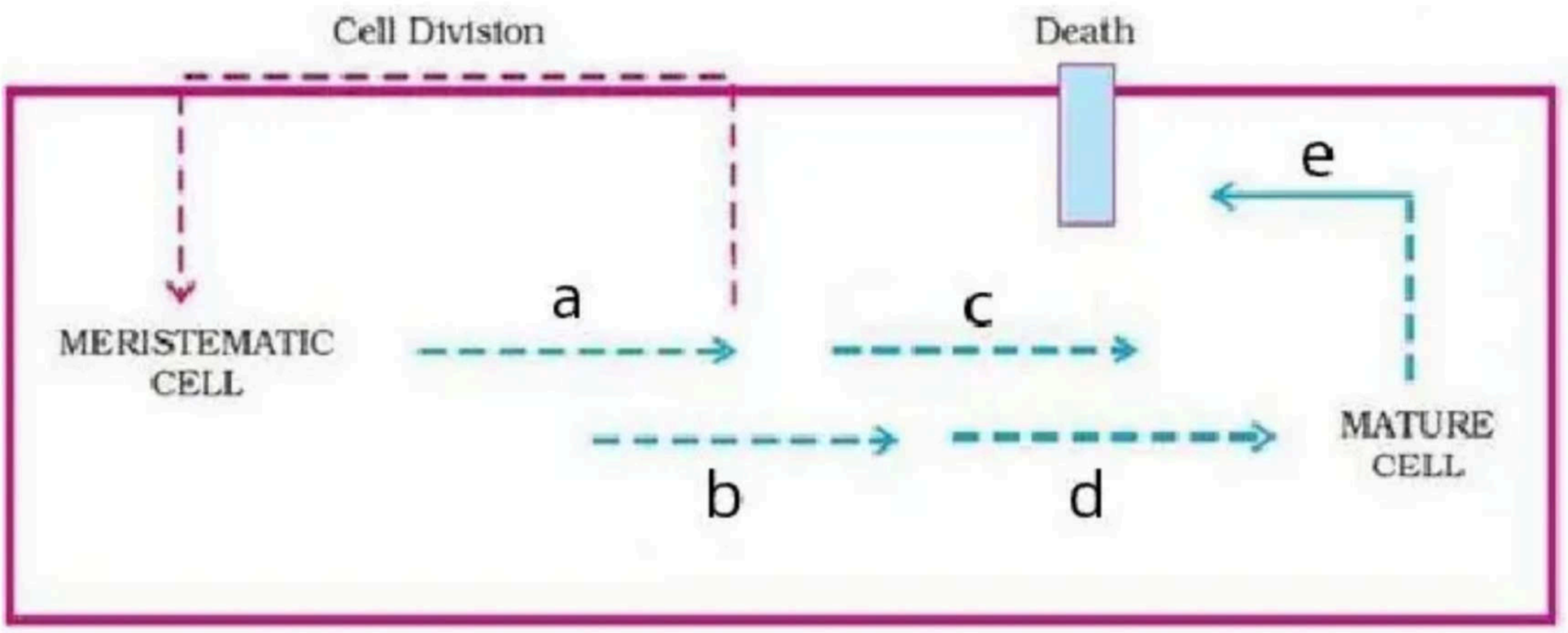
120. Development is the sum of two processes
1) Growth and cell-divison 2) Dedifferentiation and redifferentiation
3) Differentiation and dedifferentiation 4) Growth and differentiation

121. The following figure shows



1) Heterophylly in cotton, coriander and larkspur
2) Heterophylly in cotton and coriander
3) Heterophylly in larkspur
4) Heterophylly in buttercup

122. Recognise the figure and find out the correct labelling



1) b – elongation, c – differentiation, a – plasmatic growth, e – senescence, d – maturation
2) a – elongation, d – differentiation, b – plasmatic growth, e – senescence, c – maturation
3) b – elongation, e – differentation, e – differentiation, a – plasmatic growth, c – senescence, d – maturation
4) a – elongation, c – differentiation, b – plasmic growth, d – senscence, e maturation

123. Functions of auxins include
1) Use as herbiciides to kill dicot weeds 2) Inducting parthenocarpy in tomato
3) Promoting flowering in pineapple 4) All of these

124. Which accelerates the abscission in flower and fruit like thinning of cotton, cherry and walnut ?
1) ABA 2) GA 3) Ethephon 4) Auxin

125. Which of the following physiological effects is caused in plants by gibberellic acid ?
1) Rooting in stem cuttings 2) Elongation of genetically dwarf plants
3) Shortening of genetically tall plants 4) Yellowing of young leaves

126. Size of grapes increases in application of
1) Cytokinin 2) Gibberellin 3) Auxin 4) All the above

127. Ethylene is used for
1) Slowing down swelling of plant axis 2) Retarding ripening of tomatoes
3) Hastening ripening of fruits 4) Both 1 and 3

128. The most widely used compound as a source of C_2H_4 is\
1) IBA 2) Zeatin 3) Ethaphon 4) Kinetin

129. Which of the following plant hormones stimulates the germination of seed.
1) Auxin 2) Gibberellin 3) Abscisic acid 4) Ethylene

130. Which one of the PGRs would you use if you are asked to ?
a) Induce rooting in a twig b) Quickly leaf senescence
c) Delay leaf senescence
1) a-auxin, b-ABA, c-cytokinin 2) a-auxin, b-ethylene, c-GA
3) a-cytokinin, b-GA, c-auxin 4) a-cytokinin, b-ethylene, c-GA

131. Match the following

A) IAA	i) Herring sperm DNA
B) ABA	ii) Bolting
C) Ethylene	iii) Stomatal closure
D) GA	iv) Weed-free lawns
E) Cytokinins	v) Ripening of fruits

1) A – v, B – iii, C – ii, D – i, E – iv 2) A – iv, B – iii, C – v, D – ii, E – i
3) A – v, B – iii, C – iv, D – ii, E – i 4) A –iv, B – i, C – iv, D – iii, E – ii

132. The affects of apical dominance can be overcomed by which of the following hormone
1) Cytokinin 2) Gibberellin 3) IAA 4) Ethylene

133. **Assertion (A) :** The gardeners cut the top of stems of the shrubs to make them dense.
Reason (R) : The auxin found in the terminal bud inhibits growth of lateral bud.
1) Both A and R are true and R is the correct explanation of A
2) Both A and R are true and R is not the correct explanation of A
3) A is true but R is false 4) A is false but R is true

134. **Assertion (A) :** In pineapple, the auxin promote flowering.
Reason (R) : Auxins generally inhibit flowering.
1) Both A and R are true and R is the correct explanation of A
2) Both A and R are true and R is not the correct explanation of A
3) A is true but R is false 4) A is false but R is true

135. Coconut milk contains
1) Gibberellin 2) ABA 3) Auxin 4) Cytokinin

ZOOLOGY

136. The functions of the organs / organ systems in our body must be coordinated to maintain ___.
1) Haemostasis 2) Homeostasis 3) Epistasis 4) Metastasis

137. The PNS is divided into two divisons called _____ and _____ neural system
1) Central and Peripheral 2) Sympathetic and Parasympathetic
3) Cranial and spinal 4) Somatic and autonomic

138. Somatic neural system relays impulses from the CNS to _____ muscles
1) Visceral 2) Skeletol 3) Smooth 4) Involuntary

139. Visceral nervous system is the part of the _____ nervous system
1) Central 2) Peripheral 3) Somatic 4) Both 1 and 3

140. Branches that project out of the cell body also contain Nissls granule's and are called ___. These fibres transmit impulses towards the _____.
1) Axon, Cellbody 2) Nerve fibre, Gland 3) Dendrite, Soma 4) Dendrite, Axon

141. Axon's distal end is branched, and each branch terminates as a bulb like structure called ___.
1) Sympathetic knob 2) Synaptic knob 3) Synapse 4) Synaptic cleft

142. Bipolar neuron is with ___ axon and ___ dendrite, found in the ____ of eye.
1) 1, many, retina 2) 2, 2, cornea
3) Many, 1, choroid layer 4) 1, 1, retina

143. Myelinated nerve fibres are enveloped with _____ cells which formes sheath around axon
1) Schwann 2) Microglia 3) Astrocytes 4) Ependymal

144. Neurons are excitable cells because their membranes are in a _____.
1) Polarised state 2) Non polarised state 3) Depolarised state 4) Repolarised state

145. At resting state axonal membrane is comparatively more permiable to _____ ions and nearly impermiable to _____ ions respectively.
1) K^+, Na^+ 2) Na^+, K^+ 3) Na^+, PO_4^- 4) Ca^{++}, K^+

146. Ionic gradients across the resting membrane are maintained by the active transport of ions by the _____.
1) Na^+ K^+ pump 2) Na^+ Cl^- pump 3) Fe^{2+}, Mg^{2+} 4) Na^+ Fe^{2+}

147. Na-K pump transports ____ Na^+ outwards for ____ K^+ into the cell as result outer surface of axonal membrane possesses a ____ charge while its inner surface becomes _____ charged
1) 2, 3, +ve, – ve 2) 3, 2, +ve, –ve 3) 3, 2, +ve, +ve 4) 2, 3, – ve, – ve

148. Rapid imflux of Na^+ followed by the _____ of the polarity at that site is called ____.
1) Reversal, depolarization 2) Reversal, Repolarization
3) Reversal, hyper polarization 4) No change, depolarization

149. The rise in the stimulus-induced permeability to Na^+ is extremely _____ it is quickly followed by rise in permeability to _____ ions. And _____ ions diffuses outside the membrane and restores the resting potential
1) Long lived, k^+, k^+ 2) Short lived, Na^+, Na^+ 3) Long lived, Na^+, k^+ 4) Short lived, k^+, k^+

150. Impulse transmission across an electrical synapse is always ____ and electrical synapses are ___ in our system
1) Faster, Rare 2) Faster, Common 3) Slower , Rare 4) Slower, Common

151. Two cerebral hemispheres are connected by a tract of nerve fibres called
1) Corpus callosum 2) Corpora quadrigemina
3) Corpora striata 4) Corpus spongiosum

152. The layer of cells which covers the cerebral hemisphere is called
1) Cerebellar cortex 2) Cerebral medulla 3) Aorburvitae 4) Cerebral cortex

153. Fibres of the tracts are covered with myelin sheath which constitute the inner part of cerebral hemisphere is called ____
1) White matter 2) Grey matter 3) Neoplasm 4) Neopallium

154. The dorsal portion of the mid brain consists of four round swellings called as ____.
1) Corpora quadrigemina 2) Optic lobes
3) Corpora striata 4) Both 1 and 2

155. Among liver, thymus, kidney and heart, an organized endocrine structure in our body
1) Liver 2) Kidney 3) Thymus 4) Heart

156. Hypothalamus is the basal part of _____ of the forebrain
1) Diencephalon 2) Cerebrum 3) Olfactory lobe 4) Both 1 and 2

157. Synthesis and secretion of pituitary hormones is regulated by the hormones of ___
1) Cerebrum 2) Hypothalamus 3) Midbrain 4) Pons

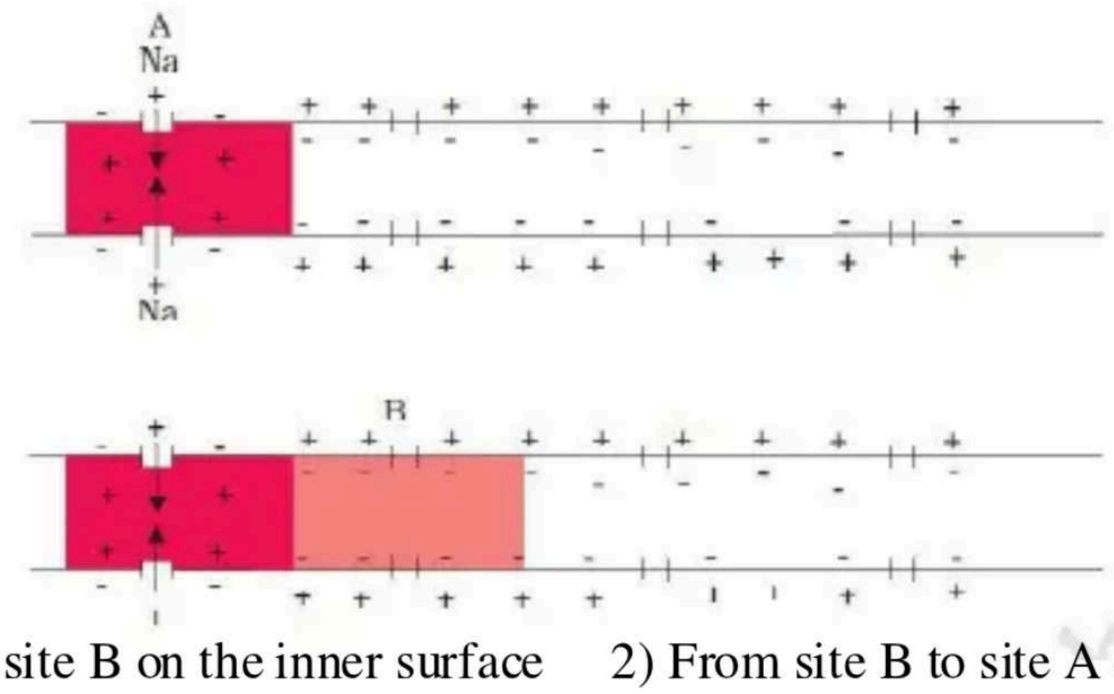
158. Hypothalamic hormone which stimulates the synthesis and release of gonadotropines is ____
1) GRH 2) GIH 3) GnRH 4) Somatostatin

159. Hypothalamic hormones reach the anterior pituitary through ____
1) Portal circulatory system 2) Lymphatic system
3) Artcrial blood circulatory system 4) Both 2 and 3

160. _____ pituitary is under the direct neural regulation of the hypothalamus
1) Anterior 2) Intermediate 3) Posterior 4) Infundibulum

161. Among ADH, MSH, PTH, PRL hormones, the one secreted by pars distalis is ___
1) PRL 2) ADH 3) MSH 4) RTH

162. During nerve impulse conduction, current flows



1) From site A to site B on the inner surface 2) From site B to site A on the outer surface
3) A to B on both sides 4) Both 1 and 2 are correct

163. Hypersecretion of GH in adults does not used further increase in height, due to the closure of _____.
1) Plate 2) Sclerotic plaque 3) Epiphyseal plates 4) Tectonial membrane

164. Synthesis and secretion of glucocorticoids from adrenal cortex is stimulated by ____
1) ACTH 2) FSH 3) LH 4) STH

165. Hormone that induces ovulation of graafinan follicle and maintains corpus luteum is ___
1) FSH 2) LH 3) PRL 4) ACTH

166. MSH which regulated pigmentation of the skin is secreted by ____ of the pituitary
1) Pars distalis 2) Pars nervosa 3) Pars intermedia 4) Pars tabularis

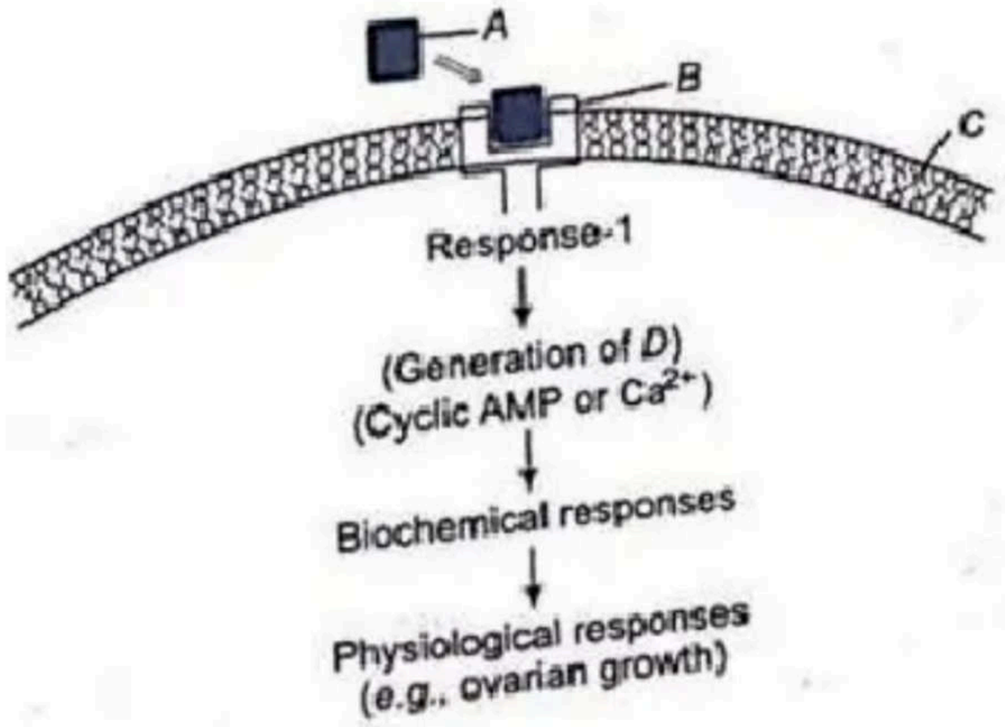
167. Hormones released by pars nervosa are actually synthesized in _____
1) Hypothalamus 2) Adenohypophysis 3) Mid brain 4) Thalamus

168. Hormone that stimulates reabsorption of water by distal tubules in the kidney is _____
1) Oxytocin 2) ADH 3) Vasopressin 4) Both 2 and 3

169. Melatonin is derived from the amino acid ____
1) Tyrosin 2) Tryptophan 3) Metheonine 4) Phenylalamine

170. Largest endocrine gland in human body is ___
1) Liver 2) Pancreas 3) Thyroid 4) Thymus

171. Identify A to D and choose the correct option.



1) A-Hormone, B-Receptor, C-Cell membrane, D-Second messenger
2) A-Hormone, B-Receptor, C-Cell membrane, D-Primary messenger
3) A-Receptor, B-Hormone, C-Cell membrane, D-Primary messenger
4) A-Receptor, B-Hormone, C-Cell membrane, D-Second messenger

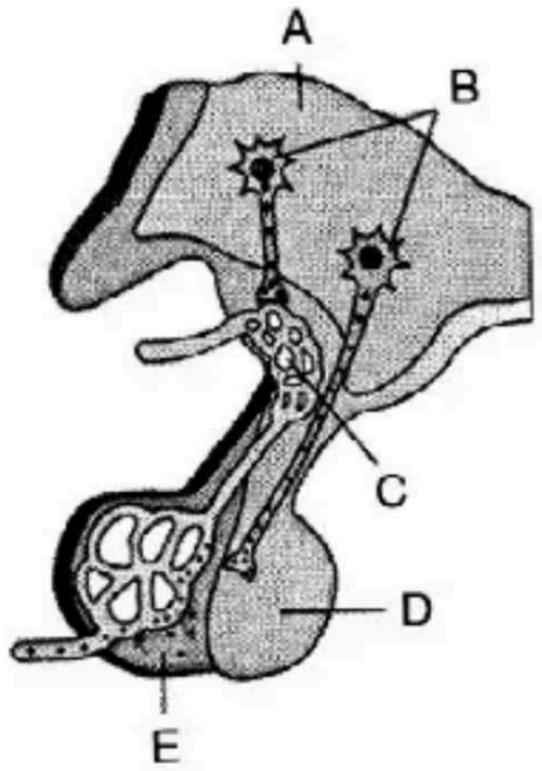
172. Hypothyroidism during pregnancy causes defective development of the baby leading to the disorder _____
1) Cretinism 2) Goitre 3) Diabetes 4) Dwarfism

173. Thyroxine is derived from the amino acid ____
1) Tryptophan 2) Tyrosine 3) Methyonine 4) Serine

174. Growth of axial hair, public hair and facial h air during puberty is stimulated by ___
1) Gluco corticoids 2) Mineralo corticoids 3) Androgenic steroids 4) Epinephrine

175. Glucagon is antagonistic to ___
1) MSH 2) Insulin 3) PTH 4) TCT

176. Identify A – E in the given figure.



1) A - Hypothalamus, B - Hypothalamic neurons, C - Portal circulation, D - Posterior pituitary, E - Anterior pituitary
2) A - Portal circulation, B - Hypothalamic neurons, C - Hypothalamus, D - Anterior pituitary, E - Posterior pituitary
3) A - Anterior pituitary, B - Portal circulation, C - Hypothalamus, D - Posterior pituitary, E - Hypothalamic neurons
4) A - Hypothalamic neurons, B - Posterior pituitary, C - Anterior pituitary, D - Portal circulation, E - Hypothalamus

177. Erythropoiethin is secreted by _____ cells of kidney
1) JG cells 2) Chromatin cells 3) MALT 4) Parafollicular cells

178. Hormones secreted by heart, kidney and gastrointestinal tract are chemically ____ hormones
1) Amine 2) Peptide 3) Protein 4) Steriod

179. $\text{Central neural system} \underset{\longleftarrow}{\overset{\text{Impulses}}{\longrightarrow}} \text{Viscera}$ (NCERT)
In which peripheral neural system involve transmition of impulses about above equation
1) Autonomic Neural system 2) Somatic Neural system
3) Sympathetic neural system 4) Visceral neural system

180. Which among the following is not a part of a system that regulate sexual behaviour, excitement, pleasure, rage, etc
1) Amygdala 2) Pons varoli 3) Hypothalamus 4) Hippocampal lobe